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(1) Research Program/Project/Study Title: R and D for Dwarf Coconut –Based Integrated Farming System in Highland of Tapaz Capiz	
Proponent Name:	Ronel V. Maximo
Academic Rank:	Instructor 1
Co-Authors (with academic rank):	Dr. Salvacion J. Legaspi – (Prof. V1) Mrs. Rocel Delfin
Campus/Satellite College/Address:	Tapaz Satellite College
Telephone/Cell Phone/Fax Number:	
E-mail Address:	

Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	1. PROFILING OF COCONUT INDUSTRY IN CAPIZ
Component Leader:	
Academic Rank:	

Component Title:	2. DOCUMENTATION OF THE GERMPLASMS COLLECTION
Component Leader:	
Academic Rank:	

Component Title:	3. DEVELOPMENT OF CROPPING SYSTEM FOR COCONUT TOWARDS A MORE SUSTAINABLE SOIL FERTILITY
Component Leader:	
Academic Rank:	

Component Title:	4. UTILIZATION OF COCONUT IN THE REPLANTING PROGRAMS FOR DENUED HILLY AREAS
Component Leader:	
Academic Rank:	

Component Title:	5. COCOLIFE: COCONUT LIVESTOCK INTEGRATED FARMING ECOSYSTEM
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☒ Natural/Biological Sciences ☐ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center
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	☒ CSRDC Crop Science Research Development Center
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(4) Site of implementation	
Municipality/City:	Tapaz
Province:	Capiz
Region:	6

(5) Classification Research: Development: ☐ Basic ☐ Pilot Testing ☒ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☒ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
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(7) Sector/Commodity Coconut, Root crops, Plantation crops, Grain crops and Legumes	(8) Discipline/Field of Specialization
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(9) Rationale Over the years, farmers have utilized nature's principle of diversity to their advantage. They have long realized that this can effectively lower their costs of production while generating higher profits. The global supply and demand of coconuts and coconut-based products have been increased tremendously over the past decades; The coconut industry in Western Visayas has potentials to become sunshine industry, but it has first to contend with threats. Francis Fegarido a Philippine coconut industry-6 officer-in charge said that the coconut farmers in the region have been thriving after the calamities that destroyed some million trees. He added that the Philippine Coconut Authority (PCA) has programs for crop rehabilitation and protection as it is battling with diseases and human factor such as; coco lisap, coco leaf beetle and the massive conversion of coconut plantations into residential and commercial areas, he said also that on top of it, is the chainsaw- which means illegal cutting of coco trees (PCA, 2019) One important practice that they have proven effective in promoting stability and diversity in the farm is intercropping - the growing of two or more crops together to increase the productivity per unit of land. Intercropping offers the best opportunity for farmers to increase their income while contributing to the maintenance of diversity of the agricultural landscape. It can be more productive and stable than growing pure crop stands. With the increased diversity in intercropping, there is also the great benefit of lower pest incidence (dela Cruz, 2009). Growing of intercrops in coconut lands produces more food and agricultural products, ensuring food security of the people in rural and urban areas. At the same time, the practice generates jobs and livelihood, enhancing farm incomes and the purchasing power of people, thus alleviating poverty in farming communities Hence, the industry has become one of the significant contributors to the economy. However, similar to the other agricultural industries, coconut has confronted by fluctuation in prices due to limited subsidy with 6 sacks of fertilizer in 1ha which could not sustained the growth of the coconut plant. Establishment of planting scheme, and good intercrop in coconut could be a possible solution to the problem and potential to coconut famers in increasing their income and minimized soil erosion.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.).

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(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Program Objectives:

1. Document the coconut industry profile in Capiz;
2. Characterize the different coconut germplasm collections in Capiz;
3. Develop a soil fertility sustainable cropping system for coconut; and
4. Develop planting scheme for hilly areas using coconut trees.

PROJECT 1: PROFILING OF COCONUT INDUSTRY IN CAPIZ (SOCIAL GROUP)

Study 1:

Objectives:

- 1.
- 2.
- 3.

PROJECT 2: DOCUMENTATION OF THE DIFFERENT GERmplasm COLLECTIONS IN TAPAZ

Study 1: Growth performance of different dwarf coconut varieties under highland of Tapaz Capiz

Objectives:

1. determine the growth performance of dwarf coconut grown under highland of Tapaz
2. find out evaluate which variety of dwarf coconut is suitable under highland of Tapaz, Capiz
3. evaluate which variety of dwarf coconut is suitable under highland of Tapaz, Capiz

Study 2: Pest and Disease resistance of the different dwarf coconut varieties under tapaz condition

1. examine the pest presence in the dwarf coconut under tapaz condition
2. evaluate the acquired resistance of different dwarf coconut against pest and diseases

PROJECT 3: DEVELOPMENT OF CROPPING SYSTEM FOR COCONUT TOWARDS A MORE SUSTAINABLE SOIL FERTILITY

Study 1: Effect of Different Root Crops as Intercrop to Newly Planted Coconuts on its Growth and Soil Fertility Dynamics

Objectives:

1. Determined the effect of different rootcrops as Intercrop of newly planted coconuts

Study 2: Grain Crops as Intercropped to Newly Established Coconuts: Its Contribution to a More Sustainable Coconut Production

Objectives:

1. evaluate the effects of grain crops as intercropped to newly establish coconut
2. Determined the potential of grain crops as intercropped to newly planted coconut and its contribution to sustainable development

Study 3: Soil Fertility Dynamics and Performance of Different Dwarf Coconut Varieties Intercropped with Different Leguminous Crops

Objectives:

1. Determined the performance of different dwarf coconut varieties Intercropped with different

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- Leguminous crops
2. Evaluate the effects of different leguminous crops on the soil fertility dynamics of

Study 4: Indigenous Rices as Intercropped to newly Established Dwarf Coconuts: Its effect on Soil Fertility Dynamics and Growth of Coconuts

Objectives:

1. determine the effect of different IR on the soil fertility dynamics of newly planted coconut
2. Evaluate the growth performance of coconut as intercrop with indigenous rice varieties
3. asses the economic potential of IR as intercrop to newly planted dwarf coconut

Study 5: Soil Fertility Dynamics and Performance of Coconut at Various Intercropping Schemes

Objectives:

1. determine the effects of various intercropping schemes on the performance of coconut
2. evaluate the influenced of different intercropping schemes on the fertility dynamics of the soil

PROJECT 4: UTILIZATION OF COCONUT IN THE REPLANTING PROGRAMS FOR DENUDED HILLY AREAS

Study 1: UTILIZATION GLYPHOSATE CONTAINING HERBICIDE AND ITS EFFECTS

Objectives:

- 1.
- 2.
- 3.

Study 2: EFFECTIVENESS OF DIFFERENT COCONUT PLANTING SCHEMES IN CONTROLLING SOIL EROSION IN THE HILLY AREAS

Objectives:

- 1.to determine the efficacy of different coconut planting schemes in controlling soil erosion
2. to evaluate the soil erosion rate of hilly areas planted with coconut
3. to established planting shemes that can control soil erosion

Study 3: PERFORMANCE OF DIFFERENT PLANTATION CROPS UNDER CAPSU-LECO PLANTING SCHEME

Objectives:

1. Determined the performance of different plantation crops under CAPSU-LECO planting schemes
2. Find out which plantation crops performed best under CAPSU-LECO planting Schemes

Study 4: SOIL EROSION POTENTIALS OF AREAS APPLIED WITH GLYPHOSATE CONTAINING HERBICIDES PLANTED WITH CORNS

Objectives:

1. determined the rate of erosion potential of areas planted with corn
- 2.Evaluate the toxicity level of areas applied with glyphosate containing herbicides planted with corn
3. Find out the interaction effect of soil erosion and application of glyphosate containing herbicide

PROJECT 5: COCOLIFE: COCONUT LIVESTOCK INTEGRATED FARMING ECOSYSTEM

Study 1: EFFECT OF STOCKING DENSITY OF DARAG AND HERITAGE CHICKEN ON THE GROWTH OF NEWLY ESTABLISHED COCONUT PLANTATION AND PASTURE GRASSES

Objectives:

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1. to determined the
- 2.
- 3.

Study 2: EFFECT OF DIFFERENT GRAZING SCHEMES ON THE PERFORMANCE OF CHICKEN, PASTURE GRASSES AND NEWLY ESTABLISHED COCONUT

Objectives:

- 1.
- 2.
- 3.

Study 3: SOIL FERTILITY DYNAMICS UNDER THE DIFFERENT GRAZING SCHEMES

Objectives:

1. Determined the effect of different grazing schemes on the soil fertility dynamics
2. evaluate which of the grazing schemes affects the fertility of the soil

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

STUDY 1: PROFILING OF COCONUT INDUSTRY IN CAPIZ



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Data to be gathered:

- 1.
- 2.
- 3.

PROJECT 2: DOCUMENTATION OF THE GERMPLASMS COLLECTION

STUDY 1: Growth Performance Of Different Dwarf Coconut Varieties In Tapaz

Experimental Site. The experiment will be conducted at coconut plantation of Capiz State University- Tapaz Satellite College, Barangay San Julian, Tapaz Capiz.

Land preparation . An area of 1.4 ha will be slash and clean ready for planting of dwarf coconut varieties

Seedling Preparation and Field Planting. Dwarf coconut varieties will be coming from the coconut breeding trial unit of Capiz State University- Burias Campus. Triangular system of planting coconut is recommended with rows laid out in a north-south direction. The distance between palms in a row should be 8 m x 8 m. Staking is done to ease in locating where the tree will be planted, dig a hole with the dimension of 50 cm diameter and 50 cm depth. Before the seedlings are placed on the hole, fertilizers are applied inside the hole as basal application

Experimental treatments and design. The experiment will be lay out to Randomized Complete Block Design (RCBD) with 8 treatments and 3 replications with 6 samples in each treatment. Treatments are designed as follows:

- T1- Aromatic**
- T2- San Isidro**
- T3- Catigan**
- T4- Talisay**
- T5- Galas**
- T6- Victoria green**
- T7- Tacunan**
- T8- Sri lanka green**

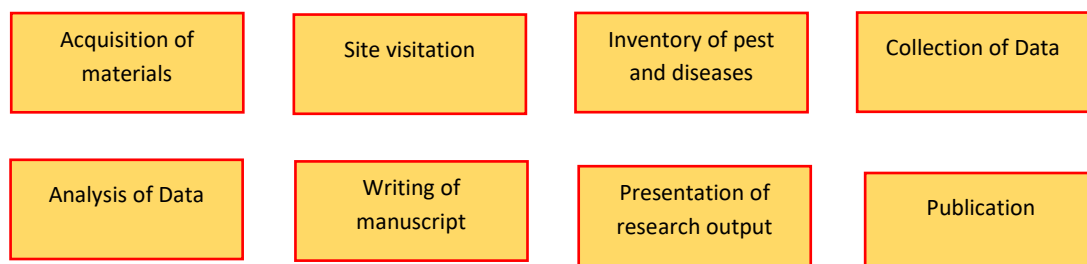
Data Gathering

Basis (Descriptors List)

Statistical Tools and Analysis

Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged in split plot in study I and single factor RCBD in study II. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.

STUDY 2: PEST AND DISEASE RESPONSES OF THE DIFFERENT DWARF VARIETIES UNDER TAPAZ CONDITION



Statistical Tools and Analysis

Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los

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Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.

PROJECT 3: DEVELOPMENT OF CROPPING SYSTEM FOR COCONUT TOWARDS A MORE SUSTAINABLE SOIL FERTILITY

STUDY 1: EFFECT OF DIFFERENT ROOT CROPS AS INTERCROP TO NEWLY PLANTED COCONUTS ON ITS GROWTH AND SOIL FERTILITY DYNAMICS

The experiment will be conducted under Newly planted coconut of CAPSU- Tapaz. An area of 1.4ha will be utilized in conducting this experiment.

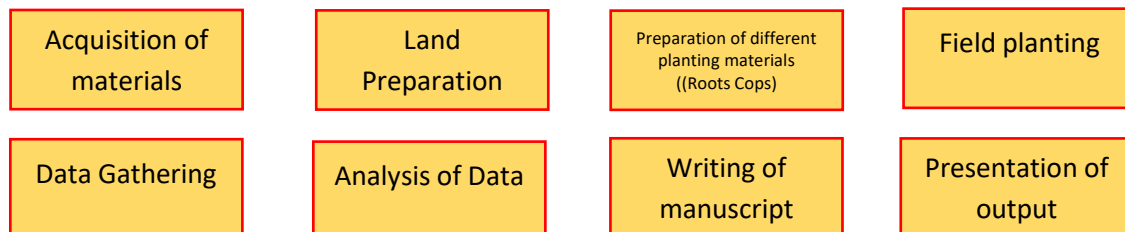
Soil sample will be collected before and after of the study. Different root crops such as sweet potato, UBE yam, Cassava and lesser yam will be collected and secure with in the area of Tapaz, Capiz.

In terms of the planting materials preparation, different root crops will be gathered with in Tapaz area. This will be planted in a direct planting method following the recommended planting distance for different root crops. Fertilizer application will be done following the recommended rate of fertilizer for different root crops.

The study will be conducted using single factor RCBD. Treatments will be composed of different root crops designated as follows: T₁ – Sweet Potato, T₂ –UBE yam, T₃- Cassava and T₄ – Lesser yam .. Each treatment will be replicated 3 times. The experimental lay-out is shown in Figure 1.0.

Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.

Activities



Treatment:

T1: Sweet Potato
T2: UBE Yam
T3: Cassava
T4: Lesser Yam

Block 1	Block 2	Block 3
T3	T2	T1
T1	T4	T3
T4	T3	T4
T2	T1	T2

Figure 1.0. experimental lay-out in RCBD

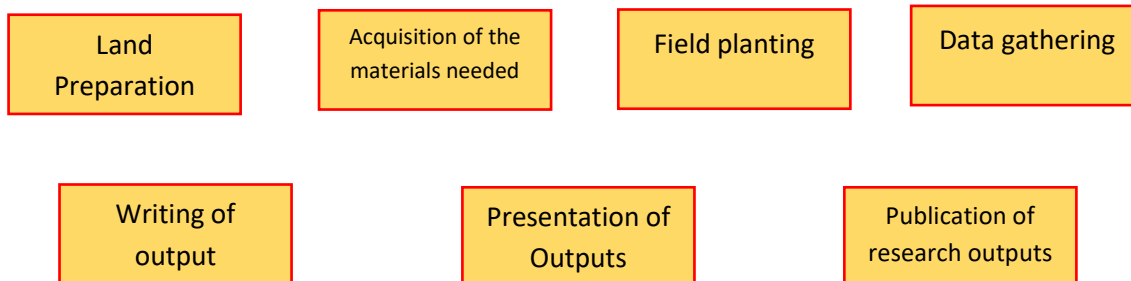
Data to be gathered:

1. Soil Analysis
2. Growth Parameters
3. Yield Parameters

STUDY 2: GRAIN CROPS AS INTERCROPPED TO NEWLY ESTABLISHED COCONUTS: ITS CONTRIBUTION TO A MORE SUSTAINABLE COCONUT PRODUCTION

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The study will be conducted at the newly established coconut project of tapaz satellite college. The experiment is a single factor experiment using the different grain crops. This includes Adlai, Corn, Soybeans, and Sorghum as intercropped. It will be carried out in a Randomized Complete Block Design with three replications. The needed materials will be collected within the locality. Planting of different grain crops will follow the standard distanced of each grains, this will be planted under newly established coconut project of CAPSU Tapaz. Land preparation will be done and Watering of the grain crops will done regularly. During the experiment, data on growth and yield of this grain crops will be collected . All data gathered will be analyzed using the analysis of variance for single factor RCBD utilizing the Statistical Tools for Agricultural Research (STAR) software.



Treatments

- T1: Adlai
- T2: Corn
- T3: Soybeans
- T4: sorghum

Data to be gathered:

1. Growth Data
2. Yield Data

Block 1	Block 2	Block 3
T2	T3	T1
T1	T1	T3
T4	T2	T4
T3	T4	T2

STUDY 3: SOIL FERTILITY DYNAMICS AND PERFORMANCE OF DIFFERENT DWARF COCONUT VARIETIES INTERCROPPED WITH DIFFERENT LEGUMINOUS CROPS

The experiment will be conducted in a single factor experiment in a Randomized complete Block Design with 4 treatments corresponding to different leguminous crops. Treatment 1 will be mungbean, Treatment 2- Peanut, Treatment 3- Soybean and Treatment 4- Peas. Preparation of experimental area will be done and soil sampling and analysis will be conducted before and after of the experiment.

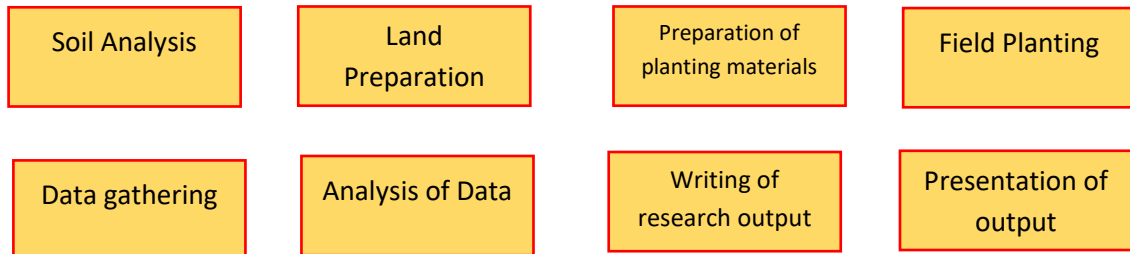
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lanting of different leguminous crops will be done by following the recommended distance of the different leguminous crops and fertilizer application will also done base on the recommended rate of the leguminous crops.

Data on growth both in coconut and legominous crops will be collected as well as soil analysis and yield parameters. Data wil be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los

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Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.



Treatment:

**T1- Mungbean
T2- Peanut
T3- Soybean
T4- Peas**

Block 1	Block 2	Block 3
T1	T2	T4
T4	T1	T3
T2	T3	T1
T3	T4	T2

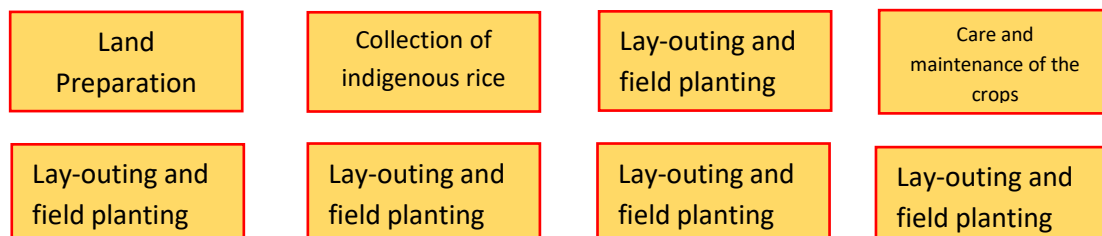
Data to be gathered:

1. Soil Analysis
2. Growth Parameters
3. Yield Parameters

STUDY 4: INDIGENOUS RICES AS INTERCROPPED TO NEWLY ESTABLISHED DWARF COCONUTS: ITS EFFECT ON SOIL FERTILITY DYNAMICS AND GROWTH OF COCONUTS

The study will be conducted using single factor RCBD. Treatments will be composed of different Indigenous Rice such as: T₁ – Mitaw, T₂ – Kalotak, T₃– Malido, T₄- Kabasag, T₅- Kutiam and T₆ – GR Violete. Each treatment will be replicated 3 times.

Growth, yield and soil analysis will be gathered and Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.



Treatment:

T1- Mitaw

Block 1	Block 2	Block 3
---------	---------	---------

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T2- Kalotak

T3- Malido

T4- KabasaG

T5- Kutiam

T6- GR- Violete

T2

T1

T5

T4

T6

T3

T6

T1

T3

T2

T5

T4

T5

T3

T1

T4

T6

T2

Data to be gathered:

1. Growth parameters

2. Yield parameters

3. Soil analysis

STUDY 5: SOIL FERTILITY DYNAMICS AND PERFORMANCE OF COCONUT AT VARIOUS INTERCROPPING SCHEMES

An area under coconut plantation will be prepared for the experiment. The study will be conducted using single factor RCBD with 5 Treatments replicated 3 times. Treatments will be composed of different Intercropping schemes designated as follows: T₁- Legumes, legumes, legumes, legumes legumes , T₂ – Legumes, grains, legumes, grains, legumes, T₃- Legume, root crops, legume, root crops, legumes, T₄- Legume, IR, Legumes, IR, Legumes and T₅- Legumes, IR, Root crops, Legumes, Legumes.

Soil analysis, growth parameters and yield parameters will be gathered and Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.

Activities

Soil Analysis

Land Preparation

Field planting and Lay-outing

Data Gathering

Analysis of Data

Writing of research output

Presentation of research output

Publication/IP Protection

Treatments :

Legumes. legumes. legumes. legumes legumes

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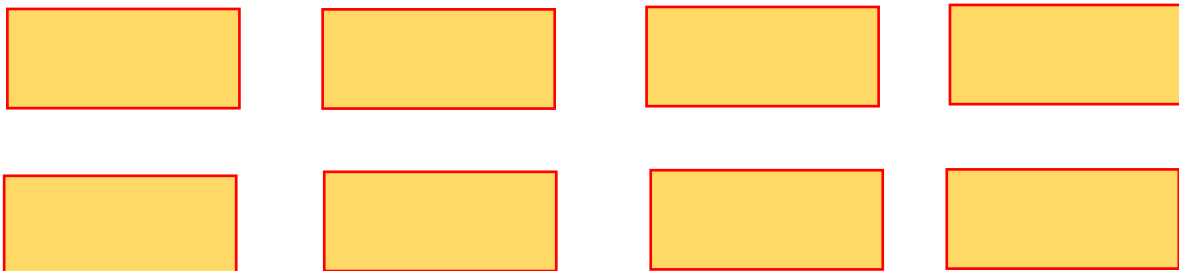
Legumes, grains, legumes, grains, legumes
 Legume, root crops, legume, root crops, legumes
 Legume, IR, Legumes, IR, Legumes
 Legumes, IR, Root crops, Legumes, Legumes

Data to be gathered:

- 1. Soil Analysis**
- 2. Growth Parameters**
- 3. Yield Parameters**

PROJECT 4: UTILIZATION OF COCONUT IN THE REPLANTING PROGRAMS FOR DENUDED HILLY AREAS

STUDY 1: UTILIZATION GLYPHOSATE CONTAINING HERBICIDE AND ITS EFFECTS (Maam Ros)

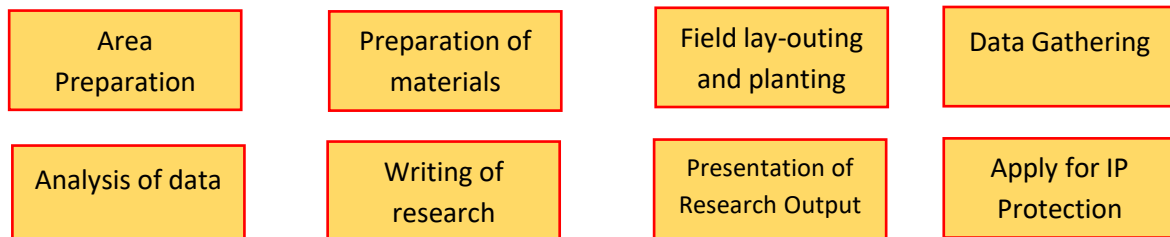


DATA TO BE GATHERED:

STUDY 2: EFFECTIVENESS OF DIFFERENT COCONUT PLANTING SCHEMES IN CONTROLLING SOIL EROSION IN THE HILLY AREAS

The study will be conducted using single factor RCBD with 3 Treatments replicated 3 times. Treatments will be composed of different Planting schemes designated as follows: T₁-Triple planting, T₂ – Hedge row plating, T₃-CAPSU LECO. Data on soil analysis, nutrient removal and erosion rate will be gathered.

Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.



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Treatments:

T1-Triple planting
T2- Hedge row planting
T3- CAPSU LECO planting Scheme

Block 1	Block 2	Block 3
T2	T3	T1
T1	T1	T3
T3	T2	T2

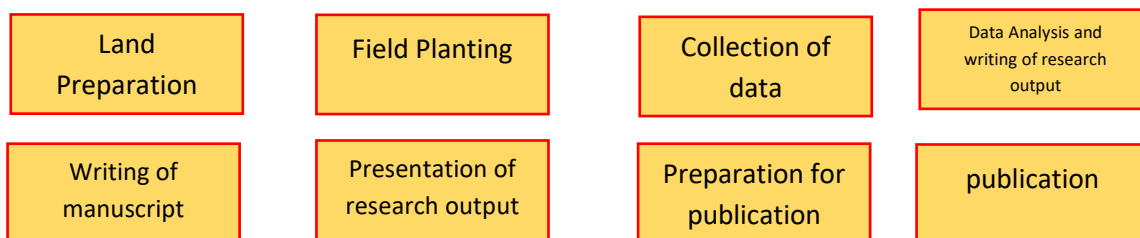
Data to be gathered:

1. – Soil Analysis
2. - Nutrient removal
3. – Erosion Rate

STUDY 3: PERFORMANCE OF DIFFERENT PLANTATION CROPS UNDER CAPSU-LECO PLANTING SCHEME

The study will be conducted using single factor RCBD. Treatments will be composed of different Plantation Crops such as: T₁ –Coffee , T₂ –Cacao, T₃- Banana and T4- Acai berry. Each treatment will be replicated 3 times.

Growth and yield parameters will be gathered and Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.



Treatment:

1. A – Coffee
2. B - Cacao
3. C - Banana
4. D - Acai Berry

Data to be gathered:

1. Growth parameters
2. Yield parameters

Block 1	Block 2	Block 3
T2	T3	T1
T1	T1	T3
T4	T2	T4
T3	T4	T2

STUDY 4: SOIL EROSION POTENTIALS OF AREAS APPLIED WITH GLYPHOSATE CONTAINING HERBICIDES PLANTED WITH CORNS

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The study will be conducted in a Randomized Complete Block Design (RCBD) with four treatments replicated three times. The treatments will be composed of Treatment 1- Corn alone, Treatment 2- Corn + Coconut, Treatment 3- Corn + Triple planting and T4- Corn + CAPSU LECO.

Data on soil erosion rate and soil analysis will be gathered and this data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance.

Field preparation
and lay-outing

Field Planting

Collection of
data

Analysis of data

Writing of
research output

Presentation of
research

Publication

Treatments:

- A – Corn alone**
- B - Corn + Coconut**
- C - Corn + Triple Planting**
- D - Corn + CAPSU-LECO**

Data to be gathered:

- 1. Soil erosion rate**
- 2. Soil analysis**

Block 1	Block 2	Block 3
T4	T3	T1
T3	T1	T3
T2	T2	T4
T1	T4	T2

PROJECT 5: COCOLIFE: COCONUT LIVESTOCK INTEGRATED FARMING ECOSYSTEM

STUDY 1: EFFECT OF STOCKING DENSITY OF DARAG AND HERITAGE CHICKEN ON THE GROWTH OF NEWLY ESTABLISHED COCONUT PLANTATION AND PASTURE GRASSES

Field preparation
and lay-outing

Field planting

Collection of
data

Analysis of data

Writing of
research output

Presentation of
research output

Publication

Treatment:

- 1. .**
- 2. .**

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3. .
4. .
5. .

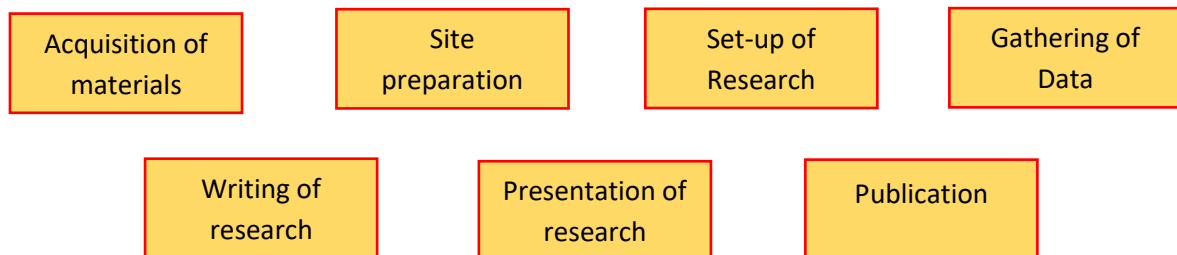
Data to be gathered:

1. .
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STUDY 2: EFFECT OF DIFFERENT GRAZING SCHEMES ON THE PERFORMANCE OF CHICKEN, PASTURE GRASSES AND NEWLY ESTABLISHED COCONUT

The study will be conducted using single factor RCBD. Treatments will be composed of different Grazing schemes such as: T₁ –free range , T₂ –partition grazing and T₃- Secluded/tunnel grazing. Each treatment will be replicated 3 times.

Growth and yield parameters will be gathered and Data will be statistically analyzed using the computer software Statistical Tool for Agricultural Research (STAR) version 2.0.1 2014, Biometrics and Breeding Informatics, Division International Rice Research Institute, Los Banos, Laguna. The presence of significant differences among treatments will be determined using Analysis of Variance (ANOVA) arranged single factor RCBD. Comparison of treatment means will be done using Duncans Multiple Range Test (DMRT) Test at 1% and 5% level of significance



**Treatments: T1- Free range
T2- Partition Grazing
T3- Secluded/Tunnel Grazing**

Data to be gathered:

1. **Growth parameters of chicken, Pasture Grasses and newly established coconut**
2. **Yield**

STUDY 3: SOIL FERTILITY DYNAMICS UNDER THE DIFFERENT GRAZING SCHEMES



Treatments:

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T1- Free range
T2- Partition Grazing
T3- Secluded/Tunnel Grazing

(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

Research Objectives	Year 1						Year 2				
	July	Aug	Sept	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May
1. Field work/observation											
2. Analysis of data gathered from observation											
3. Lay-outing, Editing and Printing of 100 copies of coffee table book											

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

(16) Targets Beneficiaries (Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

Coconut farmers, Community, Faculty and students

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(17) Personnel Requirement																																
Name	Role (Program, Project, Study Leader)	Field of Specialization																														
Percent time devoted to the Program/ Project/ Study																																
Program/Project Leader (in hours):																																
Program/Project Members (in hours):																																
(18) Budgetary Requirement																																
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